MIGUEL S. VASCO

Ph.D. Student in Computer Science

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Lisbon, Portugal

SUMMARY

As a PhD student in C.S., I am interested in understanding how can robots create multi-modal conceptual representations of their environment by interacting with a Human. How can robots infer properties of never-before seen objects from the knowledge they already possess? How can we use these concepts to teach robots how to do tasks in our homes? These are some of the questions I am interested in answering though my research!

EDUCATION

PhD in Computer Science

Técnico, University of Lisbon

09/2018 - Ongoing Lisbon, Portugal

MSc in Engineering Physics

Técnico, University of Lisbon

09/2013 - 06/2016

- Specialization: Computational Nuclear Physics
- Thesis: "3D Map of the Distribution of Metals in Cells: Application to the toxicity of Nanoparticles" - Final Grade: 19/20

Score

17.0 / 20.0

PUBLICATIONS

Online Motion Concept Learning: A novel algorithm for sample-efficient learning and recognition of human actions

International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS 2019)

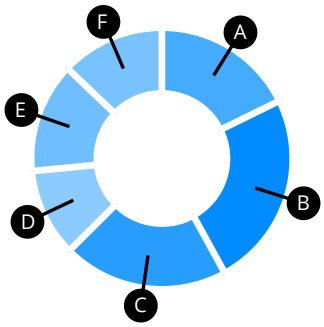
Miguel Vasco, Francisco Melo, David Martins de Matos, Ana Paiva, Tetsunari Inamura

05/2019

MOST PROUD OF

- **Brown Belt (First Kyu) in Judo**
- **Watched the sunrise on top of Mount Fuji (3776 m)**
- **Survived two Interrail trips**

MY TIME



- A Coding, Coding, Coding
- B Teaching Robots how to Think
- C Lost in Virtual Reality
- D Composing music
- E Travelling
- F Trekking

LANGUAGES

Portuguese
Native

English
Proficient

Japanese
Beginner

PUBLICATIONS

Project INSIDE: towards autonomous semi-unstructured human–robot social interaction in autism therapy

Artificial Intelligence in Medicine

Francisco Melo, Alberto Sardinha, Ana Paiva, Anabela Farias, André Mateus, Hugo Gamboa, Isabel Melo, Jhielson Pimentel, João Manuel Rodrigues, Luis Luz, Manuela Veloso, Marta Couto, Miguel Vasco, Miguel Faria, Mithun Balussery, Pedro Lima, Pedro Sequeira, Rodrigo Ventura, Rubén Solera Ureña

12/2018 URL

Publication Description

3D map distribution of metallic nanoparticles in whole cells using MeV ion microscopy

Journal of Microscopy

Miguel Vasco, Luís C. Alves, Victoria Corregidor, Andrew Bettiol, Frank Watt, Teresa Pinheiro

04/2017 DOI: 10.1111/jmi.12561

Publication Description

EXPERIENCE

Research Student

National Institute of Informatics (NII)

11/2017 - 06/2018 Tokyo, Japan

- Participated in a long-term international student internship program under the supervision of Professor Tetsunari Inamura
- Developed novel algorithm for the sample-efficient creation and recognition of human action representations in a virtual-reality environment.

Research Assistant

GAIPS (INESC-ID)

02/2017 - 06/2018 Lisbon, Portugal

- Participated in the INSIDE project, in the development of an interaction framework for robotic platforms and in outreach activities.

Private Tutor

Centro Mastermind

03/2015 - 01/2017 Lisbon, Portugal

- Helped several students achieve their desired academic results, in Physics (11º, 12º Grades) and Chemistry (11º Grade) classes.

PROJECTS

INSIDE: Intelligent Networked Robot Systems for Symbiotic Interaction with Children with Impaired Development

02/2017 - 11/2017 Porto Salvo, Portugal

http://www.project-inside.pt/

Project INSIDE explores the scientific and technological challenges involved in developing symbiotic human-robot interactions in the context of a physical game involving children with autism.

- Responsible for the development of the animation system and the decision-making state machine of the ASTRo robot.
- Responsible for the implementation of several activities to be performed in therapeutic sessions at Hospital Garcia da Horta, Almada.
- Helped in the coordination of the work done by the different research groups involved in the project
- Participated in outreach activities concerning the project.
- Participated in outreach activities concerning the project.

REFERENCES

Prof. Francisco S. Melo

(...) From the beginning, I realized that his [Miguel's] wild hair came with wild ideas. The hair never did calm down - and neither did the ideas. (...)